Team Details

Team name: [Add your team name]

a. **Team leader name:** [Add team leader name]
b. **Problem Statement:** PS 3 - Financial Health Score (AI/ML MSME Financial Health Card)
c. **Problem Statement:**

Brief about the idea

An AI/ML-driven MSME Financial Health Card that fuses alternate data (GST, UPI, Account Aggregator, EPFO) into a single, transparent, multidimensional Financial Health Score (0-100).

It turns the fragmented digital footprint of New-to-Credit / New-to-Bank MSMEs into an explainable creditworthiness view - enabling near real-time, inclusive and prudent lending.

Designed to plug into India's ULI / OCEN / Account Aggregator rails and into the bank's mobile / loan-origination stack.

Outcome: fewer rejected viable borrowers, faster financial inclusion, and better portfolio quality.

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

How it is different

Fuses 4 alternate-data sources into ONE score vs siloed point checks

Dual-engine: transparent glass-box pillar score + ML challenger (not a black box)

Per-applicant SHAP explainability - audit-ready for regulated lending

How it solves the problem

Scores credit-invisible MSMEs that lack audited financials; expands inclusion

Near real-time re-scoring on fresh consented data

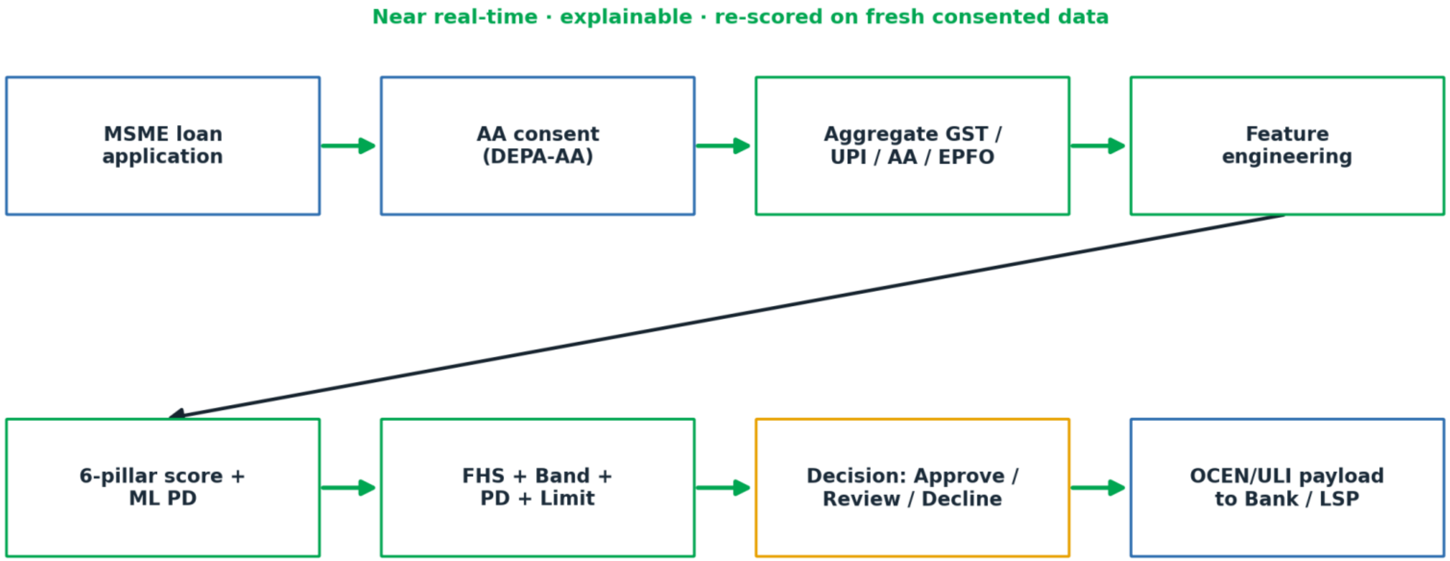
USP

Explainable-by-design + interoperable (OCEN/ULI) + inclusion-first, deployable into the bank's stack

List of features offered by the solution

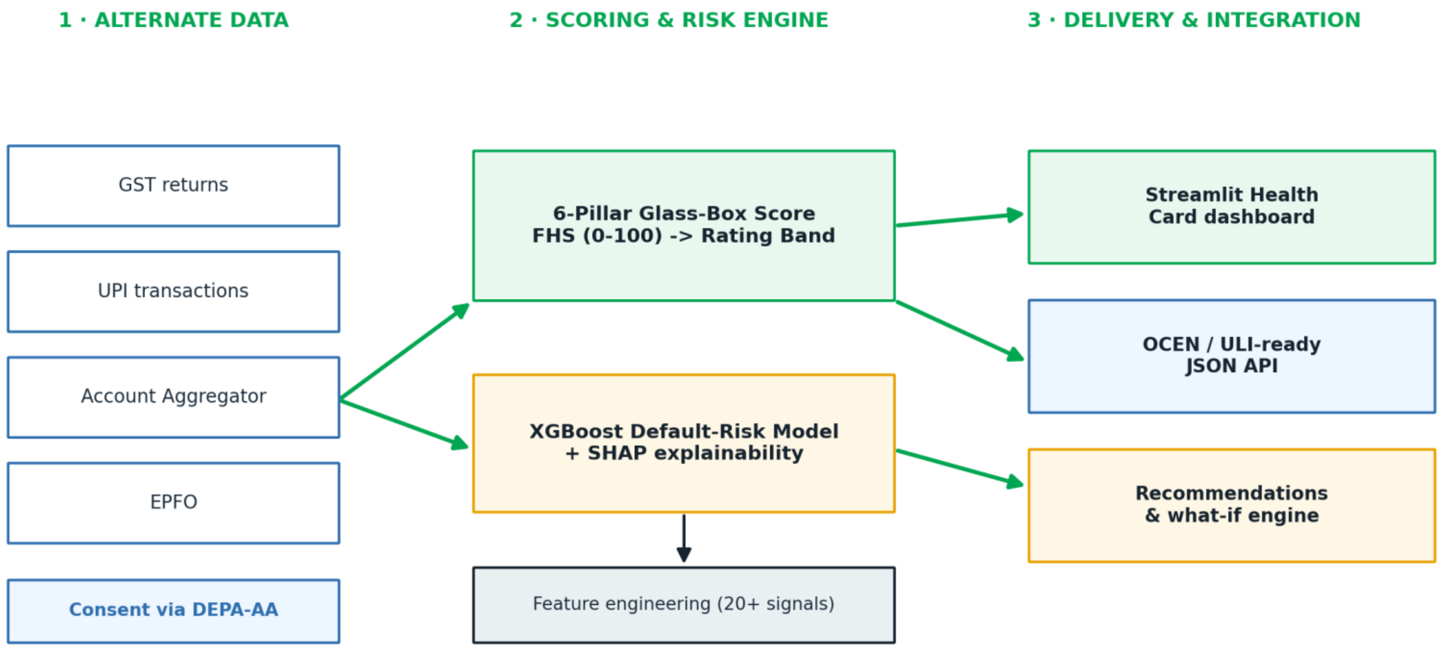
- 6-pillar Financial Health Score (0-100) with rating band (AAA-C)
- Alternate-data aggregation: GST, UPI, Account Aggregator, EPFO
- Dual PD: transparent rule-based + XGBoost ML (AUC ~ 0.78, KS ~ 0.49)
- Per-applicant SHAP explainability - what drives each borrower's risk
- Indicative credit-limit recommendation
- What-if simulator for live re-scoring and RM coaching
- Portfolio & financial-inclusion analytics + CSV export
- OCEN / ULI-ready JSON API with DEPA-AA consent framing
- Score-improvement recommendations for the MSME

Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution (optional)

Architecture diagram of the proposed solution



Technologies to be used in the solution

App & visualisation

Python, Streamlit, Plotly

Machine learning & data

scikit-learn, XGBoost, native SHAP tree contributions, pandas, NumPy

Scoring engine

Transparent weighted 6-pillar model (glass-box, auditable)

Integration & interoperability

OCEN / ULI JSON payload, Account Aggregator (DEPA-AA) consent framing

Deployment

Streamlit Community Cloud (live), GitHub, Docker-friendly, sandbox-API ready

Estimated implementation cost (optional)

MVP / POC: near-zero - open-source stack + free cloud tier (already live)

Production (indicative, variable with volume):

Containerised hosting / compute for the scoring microservice

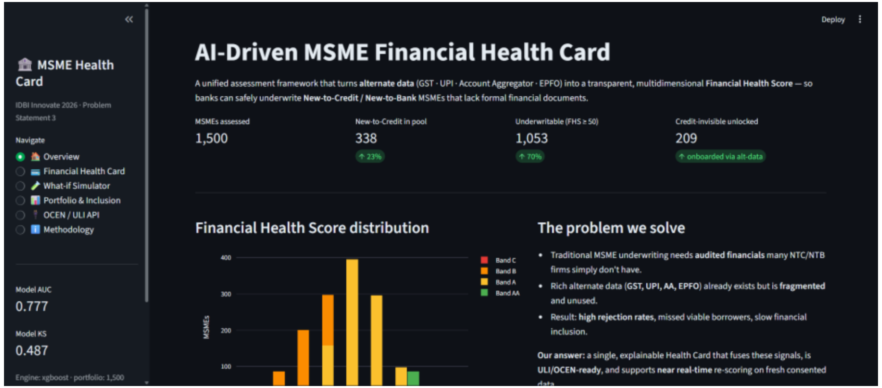
Account Aggregator data-fetch fees (per consent pull)

GST / bureau API costs (per hit)

Model ops, monitoring & audit logging

Marginal cost per assessment is low -> strong unit economics for inclusion lending at scale.

Snapshots of the prototype



Prototype Performance report/Benchmarking

ML default-risk model: AUC ~ 0.78, KS ~ 0.49 (KS > 0.4 = strong in credit scoring)

Demo portfolio: 1,500 MSMEs scored; ~23% New-to-Credit

~70% underwritable (FHS >= 50); 200+ credit-invisible MSMEs unlocked via alt-data

Instant scoring latency; fully reproducible, transparent pipeline

vs baseline structured-data-only default models (~16-22% accuracy per the bank's problem) -> large uplift PLUS regulator-friendly explainability

Additional Details/Future Development (if any)

Live sandbox integrations: Account Aggregator, GST, UPI (NPCI), EPFO

Bureau-data fusion, reject-inference, and bias / fairness monitoring

Sector-specific scorecards and seasonality-aware models

Vernacular RM co-pilot and straight-through auto-decisioning within limits

Direct embed into IDBI mobile app / LOS; ULI / OCEN production onboarding

Continuous re-scoring and early-warning signals for portfolio quality

Provide links to your:

GitHub Public Repository: https://github.com/rtm20/MSME-Health-Card

- **Demo Video Link (3 Minutes):** [Add your 3-min demo video URL]
- **Final Product Link:** https://rtm20-msme-health-card-app-ar3kpv.streamlit.app/
- Final Product Link

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Thank you!